MIL-P-43331A

11 September 1969
SUPERSEDING
MIL-P-43331(GL)
16 April 1965

MILITARY SPECIFICATION

PAPER, MANILA

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers rope manila paper and chemical wood manila paper and reflects item requirements parallel to the specifications of the Joint Committee on Printing.

* 1.2 Classification.- Manila paper shall be of the following types, as specified:

Type I - Rope manila (JCP M20)

Type II - Chemical wood manila (JCP M10)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-25 - Printing and Stationery Paper, Packaging and Packing of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9310

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* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

JOINT COMMITTEE ON PRINTING PUBLICATIONS:

GOVERNMENT PAPER SPECIFICATION STANDARDS

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI) PUBLICATIONS

T 402 m-49 - Conditioning Paper and Paperboard for Testing.

T 401 m-60 - Fiber Analysis of Paper and Paperboard.

T 410 os-68 - Weight Per Unit Area (Basis Weight or Substance) of Paper and Paperboard.

T 409 os-61 - Machine Direction of Paper.

T 423 m-50 - Folding Endurance of Paper.

T 423 su-68 - Folding Endurance of Paper (Schopper Test).

T 435 m-52 - Hydrogen Ion Content (pH) of Paper Extracts.

(Copies of TAPPI testing methods may be obtained from the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, New York 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard sample.- Color, finish, formation and cleanliness shall conform to the standard sample as adopted by the Joint Committee on Printing.

3.2 Manila paper.- Type I and type II manila paper shall conform to all the requirements set forth in the Joint Committee on Printing Standards JCP M20 and JCP M10 respectively (see 6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspections.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

* 4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item.- The end item shall be examined in accordance with the defects set forth in the applicable paragraphs at the inspection levels and acceptable quality levels (AQL's) in table I. Random samples shall be drawn from each lot of the end item for examination of visual, dimensional, count and preparation for delivery defects. The lot size for the purpose of determining the sample size, in accordance with MIL-STD-105, shall be stated in table I.

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TABLE I. - Sampling procedure

Examination paragraph	End item	Lot size in unit of	Sample unit	Inspection level	AQL	Defects in 100 units (DHU) of
4.2.2.1.1	8 by 10-1/2 in. sheet	Reams	One 8 by 10-1/2 in. sheet	S-4	4.0	8 by 10-1/2 in. sheets
	Larger than 8 by 10-1/2	Reams	One 8 by 10-1/2 in. sheet	S-2 and S-4 1/	4.0	8 by 10-1/2 in. sheets
4.2.2.1.2	All size sheets	Reams	Full size sheet	S-4	4.0	Sheets
4.2.2.1.3	All size sheets	Reams	Full size sheet	S-2	2.5	Sheets
4.2.2.1.4	All size sheets	Reams	Ream or quantity between markers (when skid packed)	S-2	---	---
4.2.2.1.5	All size sheets	Shipping containers	Shipping container	S-2	2.5	Shipping containers
4.2.2.1.6	All size sheets	Shipping containers	Shipping container	S-2	2.5	Shipping containers

1/ For sheets larger than 8 by 10-1/2 inches, full size sheets shall be selected at the S-2 level and cut into as many sheets as possible, 8 by 10-1/2 inches and randomly arranged for selection of the examination samples at the S-4 level.

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4.2.2.1.1 Examination of the end item for color, finish, formation and cleanliness.- Only one sheet, 8 by 10-1/2 inches or larger, selected in accordance with table I shall be taken from a package. Both sides of the sample unit shall be examined.

Examine	Defects
Appearance	Cleanliness, finish or formation (as applicable) does not compare favorably with standard sample.
	Color not uniform on both sides; does not compare favorably with the standard sample.
	Presence of holes, dirt or oil spots, lumps, foreign or undigested matter, slivers, shives, scales, shiners, scabs, slime spots or mottle larger than or more than in standard sample.

4.2.2.1.2 Examination of the end item for workmanship.- No more than one sheet, selected in accordance with table I, shall be taken from a ream.

Examine	Defects
Workmanship	Sheets not trimmed clean; ragged, torn, uneven or crushed edges.
	Paper curls, does not lie flat, i.e., not flat after reasonable flexing.
	Any tear, cut, wrinkle, fold, chafed or scuff mark.
	Presence of lint, fuzz or any particles which will pick, lift, fluff or pile on the press blanket, applicable to type II only.

4.2.2.1.3 Examination of the end item for dimensional defects.- No more than one sheet selected in accordance with table I, shall be taken from a ream.

Examine	Defects
Sheet size	Less than specified in either direction.
Sheets 8-1/2 by 14 inches or less	Oversize by more than 1/16 inch in either dimension from size specified.

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Examine	Defects
Sheets over 8-1/2 by 14 inches	Oversize by more than 1/6 inch in either dimension from size specified.
Squareness	Difference between the length of two diagonals exceeds 1/32 inch for sheets 32 inches or less in length, or exceeds 1/16 inch for sheets over 32 inches in length.

4.2.2.1.4 Examination of the end item for count.- The average count of sheets per package or quantity between markers, when paper is skid packed, shall be not less than the specified quantity.

4.2.2.1.5 Examination of the end item for color uniformity.- Color not uniform within the unit package and throughout the skid or shipping container.

4.2.2.1.6 Examination of preparation for delivery.- An examination shall be made to determine that the requirements of Section 5 are met. Shipping containers, fully packed shall be selected prior to the closing operation. Close shipping containers shall be examined for closure defects.

Examine	Defects
Markings (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling.

* 4.2.3 Testing of the end item.- The end item shall be tested for the characteristics in 3.3 for each lot of material and in accordance with instructions in table II below. The sample unit shall be 1000 square inches of paper. The lot size shall be expressed in units of individual packages of sheets; no more than one sample unit shall be drawn from any one package. The inspection level, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be S-1 and the acceptable quality level (AQL), shall be 2.5 expressed as defects per one hundred sample units.

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INSTRUCTIONS FOR TESTING OF THE END ITEM

* TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per sample unit	Results Reported As		Inspection Level	AQL
	Requirement	Test Method	Individual Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Conditioning for testing	3.2	T402m-49			All testing				
Stock	3.2	T401m-60		X	2 on composite 1/		5 percent		
Basis weight	3.2	T410os-68		X	1 (Average of 1000 square inches)		0.1 pound		
Grain direction	3.2	T409os-61			1	X	---		
Folding Endurance									
MIT Double Folds	3.2	T423m-50	2/	X	10		Double fold		
Schopper Double Folds	3.2	T423su-68		X	10		Double fold		
Acidity (Type I only)	3.2	T435-68		X	2 on composite		0.1 pH unit		

1/ Composite shall consist of small pieces of equal size taken from each sample unit.

2/ The MIT Fold test, deleted in T 423 su-68, as of December, 1968, will be reissued by TAPPI at a later date. Use T 423 m-50 pending its reissue.

AMXRL Form 598
8 February 1965

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2.)

* 5.1.1 Level A.- The manila paper shall be packaged in accordance with the applicable level A requirements for printing paper as indicated in table III of this specification.

* 5.1.2 Level C.- The manila paper shall be packaged in accordance with the applicable level C requirements as specified in PPP-P-25.

* 5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- The manila paper, packaged as specified in 5.1, shall be packed in accordance with the applicable level A requirements for printing papers as specified in PPP-P-25 and as indicated in table III of this specification.

5.2.2 Level B.- The manila paper, packaged as specified in 5.1, shall be packed in accordance with the applicable level B requirements for printing papers as specified in PPP-P-25 and as indicated in table III of this specification.

5.2.3 Level C.- The manila paper, packaged as specified in 5.1, shall be packed in accordance with the applicable level C requirements for printing papers as specified in PPP-P-25.

* 5.3 Marking.- In addition to any special marking required by the contract or order, interior packages, shipping containers, and skids of paper (when applicable) shall be marked in accordance with the applicable marking requirements of PPP-P-25.

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TABLE III.- Packaging and packing of manila paper.

Ream						Shipping container			
Nomenclature	Size (in.)	Wt. (lb.)	Approximate Thickness (in.)	Wrapped	Ream marked w/marker placed lengthwise	Approximate inside length and width (inside of liner) (in.)	Quantity	Approximate net wt. (lb.)	Type
Type I	16x21	31.11	3.45	No	Yes	21 x 16	6 reams	197	Fiberboard
	32x42	124.56	3.45	No	Yes	42 x 32	2 reams	250	Fiberboard
Type II	16x21	27.22	3.20	No	Yes	21 x 16	6 reams	164	Fiberboard
	32x42	108.88	3.20	No	Yes	42 x 32	2 reams	218	Fiberboard

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6. NOTES

6.1 Intended use.

6.1.1 Type I.- Type I paper is intended to be used for face and back letterpress printing of file folders, tickets, tags, cards, etc., requiring toughness and durability. Printed matter may include text and line illustrations.

6.1.2 Type II.- Type II paper is intended to be used for face and back letterpress and offset printing of light file folders, tickets, tags, cards, and minimum quality covers for books and pads, etc. Printed matter may include text and line illustrations.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required (see 1.2).

(c) Grain direction (see 3.1).

(d) Pressroom conditions (see 3.2).

(e) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard comparison samples may be obtained upon application to the procuring activity issuing the invitation for bids.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Project No. 9310-0065

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Paper, Manila MIL-P-43331A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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